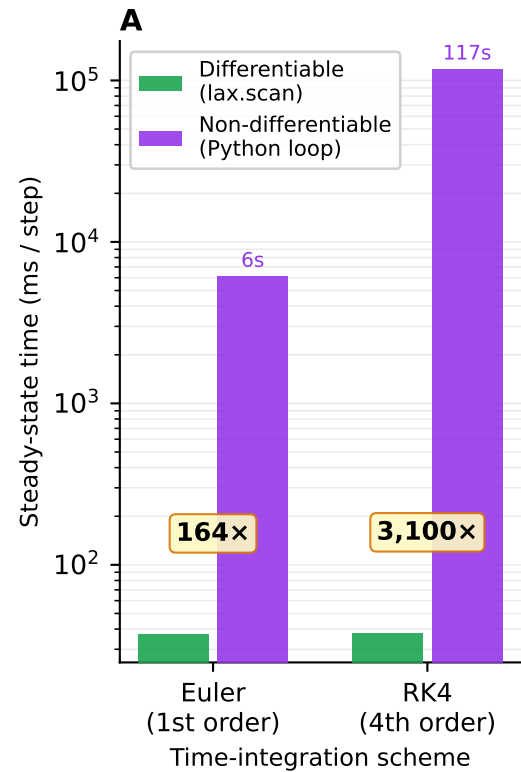
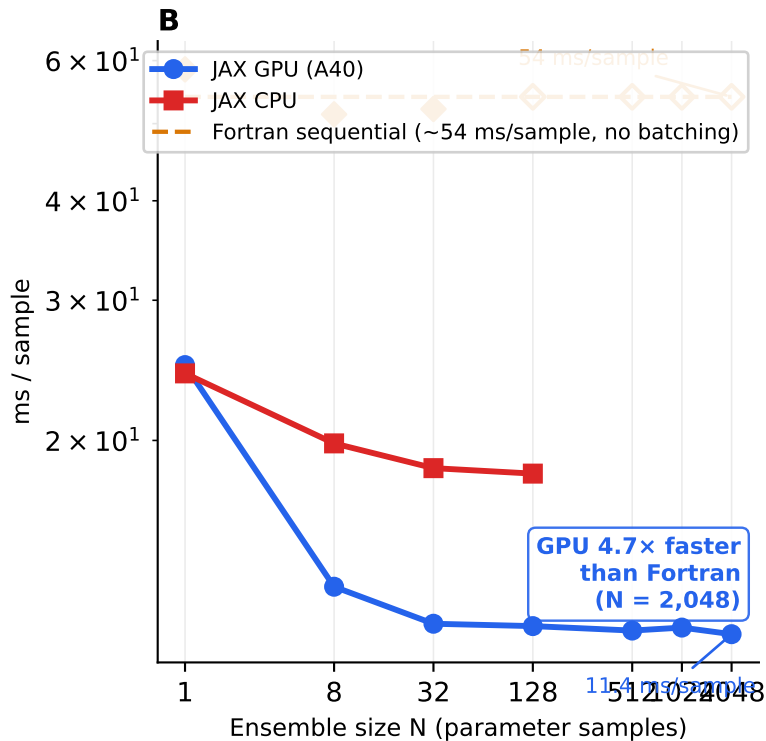


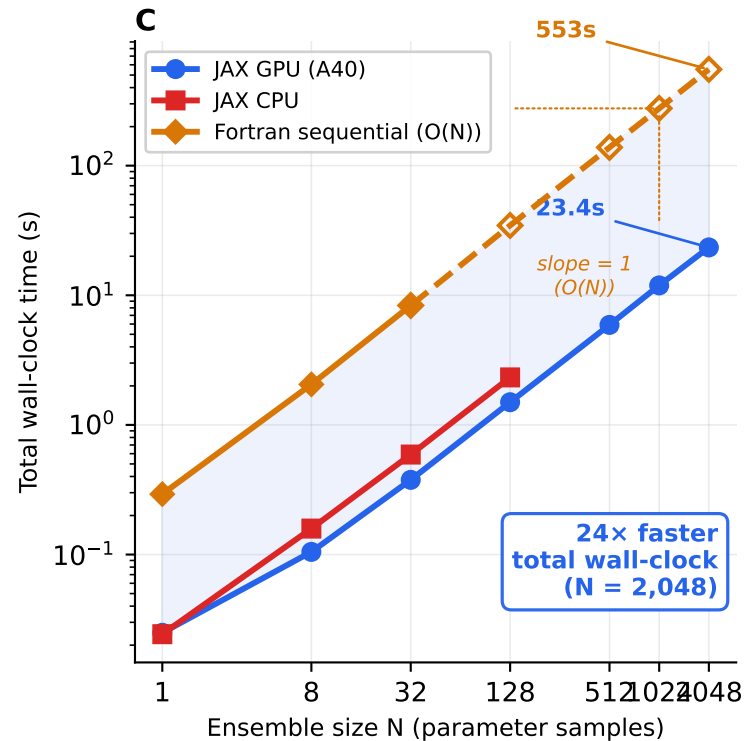
CLM-ML-JAX: differentiable multi-layer canopy physics on GPU
 CHATS7 (46-layer canopy column, May 2007) · Ensemble: 5-parameter perturbations, Euler timestepping, A40 GPU



Gradient computation via lax.scan eliminates Python-loop overhead



JAX amortizes cost across samples via vmap;
 Fortran must execute each sample serially



Fortran scales O(N); JAX GPU wall-clock grows sub-linearly via vmap parallelism